Cozy Bear Cottage （小熊中文） — Build Instructions

How to create the React web app · Meta Global AI Developer Hackathon · Build window Oct 26 – Nov 4, 2026 · Final submission Nov 4, 9:00 PM ET

What the final product is

A **mobile-first web app** (React + Vite, installable as a PWA): a parent-teacher opens a link on any phone, tablet, or laptop — no app store, no install required. Three features: Lesson Builder (Muse Spark), Living Flashcards (Muse Image), Point-and-Say (SAM 3.1).

What is needed before coding

- **Meta Model API access** — granted to selected hackathon participants, with $150 in API credits. Access to Muse Spark, Muse Image, and SAM 3.1 comes through this API. *This is why the application (due Oct 25) comes first.*
- **Node.js 18+** and a code editor; **Muse Code** (Meta's coding agent, provided to participants) for the build.
- **GitHub account + repo** for the project code (also linked in the application/submission).
- **Hosting account** — Vercel, Netlify, or Cloudflare Pages (free tier is enough). Deploys automatically from the GitHub repo.
- Your design assets are ready: the white bear (reading) mascot, Vivian's artwork and palette, in the project files.

Step 1 — Scaffold the app

```
npm create vite@latest cozy-bear-cottage -- --template react
cd cozy-bear-cottage
npm install
npm install vite-plugin-pwa        # installable, full-screen on phones
npm run dev                        # local preview
```

Step 2 — The API proxy (required)

The browser must never hold the Meta API key. Add one small serverless function (Vercel/Netlify function, ~40 lines) that receives requests from the app, attaches the key from an environment variable (`META_API_KEY`), calls the Meta Model API, and returns the result. All three features call the models only through this proxy.

Step 3 — Lesson Builder (Muse Spark)

- Form: age range (3–4 / 5–6) + theme (colors, animals, numbers, family, food, body, transport, nature).
- Proxy call to Muse Spark with a structured prompt: return lesson JSON — opening song, word set (character + pinyin + English), a 3-line story, two games, closing song, bilingual teaching script.
- Render the JSON as a clean, readable 45-minute lesson page the teacher can follow.

Step 4 — Living Flashcards (Muse Image)

- For each word in the lesson, call Muse Image with a locked style prompt: the same little white bear teacher, soft storybook style, consistent on every card.
- Render a flashcard grid; a print stylesheet (`@media print`) turns it into printable flashcards, trace-the-character worksheets, and coloring pages via the browser's Print → Save as PDF.

Step 5 — Point-and-Say (SAM 3.1)

- Camera via `getUserMedia`, with photo upload as a fallback.
- The child taps an object → send the image + tap coordinates through the proxy to SAM 3.1 → receive the object's mask/outline; highlight it on screen.
- Send the cropped object to Muse Spark to name it: Chinese character + pinyin + English.
- Speak it with the Web Speech API: `speechSynthesis` with a `zh-CN` voice. The child repeats it; any brave attempt earns a star.
- If the object matches one of this week's target words, celebrate it specially ("You found 红色！").

Step 6 — Deploy

```
git push            # to the GitHub repo
# Vercel/Netlify imports the repo → builds → gives a public URL
# Set META_API_KEY in the host's environment settings
```

The public URL is what goes into the final submission and what judges open in the demo.

Build order (matches the plan in the application packet)

- **Oct 26–27:** Scaffold + proxy + Lesson Builder end-to-end (colors theme first).
- **Oct 28–29:** Living Flashcards — style lock-in + print export.
- **Oct 30–Nov 1:** Point-and-Say — segmentation, naming, speech, stars.
- **Nov 2:** Real-world test — generate the actual Saturday lesson and use it.
- **Nov 3:** Polish, fallbacks (photo upload if camera is denied), record the demo.
- **Nov 4:** Final submission before 9:00 PM ET.

State stays simple: themes, lessons, and word sets as JSON — no database needed for the MVP. Design: light-blue theme and rainbow accents from Vivian McDonald's artwork; visual design credited to Vivian McDonald.